Clustering and Initialization:
An Experimental Comparison of Random K-means and K-means++

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Abstract

This report investigates the effect of centroid initialization on K-means clustering. Random initialization and K-means++ are implemented from scratch in Python and compared using a synthetic Gaussian dataset and the handwritten digits dataset. Each method is evaluated across 20 trials using inertia and its variability. K-means++ achieves greater consistency on the synthetic dataset. On handwritten digits, random initialization achieves slightly lower mean inertia, while K-means++ exhibits lower variability. These findings highlight the importance of dataset characteristics when assessing initialization methods.

1 Introduction

Clustering is an unsupervised machine learning task that aims to identify groups of similar observations without relying on predefined class labels. K-means is a widely used clustering algorithm because it is relatively simple and computationally efficient. It partitions observations into a predefined number of clusters by minimizing the sum of squared distances to cluster centroids.

An important limitation of K-means is its sensitivity to the initial selection of centroids. When starting centers are selected randomly, different initializations can produce different final cluster assignments and objective values. This occurs because the iterative algorithm may converge to local rather than global minima.

Arthur and Vassilvitskii [1] proposed K-means++, an initialization strategy that encourages the selection of well-separated starting centers. Rather than choosing every centroid uniformly, the method assigns higher selection probabilities to observations farther from previously selected centers.

The objective of this project is to compare random K-means initialization with K-means++ in terms of final inertia and stability across repeated trials.

Both methods are implemented from scratch using Python and NumPy. The experiments consider two datasets: a synthetic two-dimensional Gaussian dataset with three clusters, and a real-world dataset of handwritten digits represented by 64 numerical features.

Each initialization method is evaluated across 20 trials per dataset. This allows the comparison of average clustering performance, variability, and the best observed solutions.

2 Methodology

Both initialization methods use the same iterative clustering procedure. Their main difference is how the starting centroids are selected.

2.1 Standard K-means

Let $X = \{x_1, x_2, \ldots, x_n\}$, where $x_i \in \mathbb{R}^d$, be a dataset containing n observations.

K-means seeks K clusters by minimizing the within-cluster sum of squared distances:

$$J = \sum_{i=1}^{n} \|x_i - \mu_{c_i}\|_2^2 , \tag{1}$$

where μ_{c_i} is the centroid assigned to observation x_i.

The implemented algorithm follows three main steps.

Initialization. For random initialization, K distinct observations are sampled uniformly without replacement. These observations become the initial centroids.

Cluster assignment. Each observation is assigned to its closest centroid:

$$c_i = \underset{j \in \{1,\dots,K\}}{\operatorname{argmin}} \|x_i - \mu_j\|_2^2 . \tag{2}$$

NumPy broadcasting is used to calculate squared Euclidean distances, and `np.argmin` identifies the nearest centroid.

Centroid update. Each centroid is updated to the mean of its assigned observations:

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i, \tag{3}$$

where C_j represents the observations assigned to cluster j.

The assignment and update steps are repeated until the Euclidean movement of every individual centroid is below 10^{-4}, or a maximum of 100 iterations has been reached.

Because K-means can converge to different local minima, initialization may affect its final clustering cost.

2.2 K-means++ Initialization

K-means++ was proposed by Arthur and Vassilvitskii [1] to improve the choice of starting centroids.

The procedure is:

1. Select the first centroid uniformly at random.
2. Calculate the distance from each observation to its nearest already-selected centroid.
3. Select the next centroid with probability proportional to the squared distance.
4. Repeat until K centroids have been selected.

Let C denote the set of selected centroids and define

$$D(x) = \min_{\mu \in C} \|x - \mu\|_2. \tag{4}$$

The probability of selecting observation x is

$$P(x) = \frac{D(x)^2}{\sum_{x' \in X} D(x')^2}. \tag{5}$$

This procedure is known as D^2-weighted sampling.

Observations that are farther from the selected centroids have a greater probability of being chosen. However, the selection remains probabilistic.

After initialization, K-means++ uses the same clustering steps as standard K-means.

The original study establishes theoretical guarantees on the expected initialization cost, but does not imply better final inertia in every individual run.

2.3 Evaluation Metrics

The primary metric is final **inertia**, defined by Equation 1.

A smaller inertia indicates a more compact clustering solution according to the K-means objective.

For T trials, mean inertia is calculated as

$$\bar{J} = \frac{1}{T} \sum_{t=1}^{T} J_t. \tag{6}$$

The population standard deviation is calculated as

$$\sigma_J = \sqrt{\frac{1}{T} \sum_{t=1}^{T} (J_t - \bar{J})^2}. \tag{7}$$

A smaller standard deviation indicates less variation in the final inertia across trials.

The best observed inertia is

$$J_{\text{best}} = \min_{t=1,\ldots,T} J_t. \tag{8}$$

Since inertia depends on dataset size, dimensionality, and scale, raw values are compared between methods within the same dataset.

3 Datasets

3.1 Synthetic Gaussian Dataset

The synthetic dataset contains 150 two-dimensional observations, generated using NumPy.

Three Gaussian groups were created, each with 50 observations and identity covariance:

$$\begin{aligned} X_1 &\sim \mathcal{N}((2,2), I_2), \\ X_2 &\sim \mathcal{N}((8,8), I_2), \\ X_3 &\sim \mathcal{N}((2,8), I_2). \end{aligned} \tag{9}$$

The dataset was generated with random seed 42. The number of clusters was set to $K = 3$.

The separation between the groups allows the effects of initialization to be examined visually.

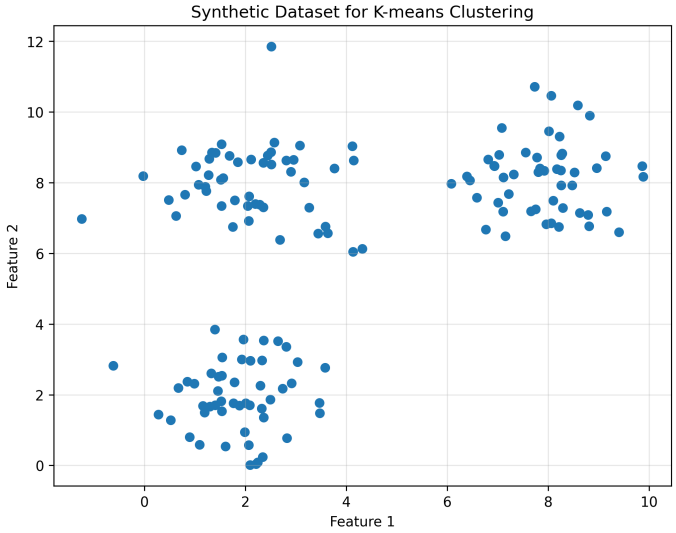


Figure 1: Synthetic dataset with three Gaussian groups and 150 observations.

3.2 Handwritten Digits Dataset

The handwritten digits dataset was loaded using `load_digits()` from scikit-learn [2].

It contains 1,797 grayscale images of digits from 0 to 9.

Each image is 8×8 pixels, represented as a vector of 64 numerical features:

$$X_{\text{digits}} \in \mathbb{R}^{1797 \times 64}. \tag{10}$$

The number of clusters was set to $K = 10$.

Although true digit labels are available, they were not used during clustering.

This dataset is more complex than the synthetic example because it has higher dimensionality and overlapping digit patterns.

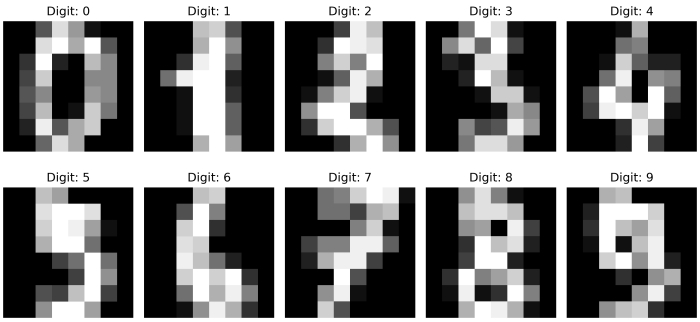


Figure 2: Examples of handwritten digit images from the Digits dataset.

4 Experimental Setup

4.1 Implementation

K-means and K-means++ were implemented from scratch using Python and NumPy.

Separate functions perform centroid initialization, cluster assignment, centroid updating, and inertia calculation.

Scikit-learn was used only to load the real-world dataset. Matplotlib was used for visualization.

4.2 Configuration

Table 1 summarizes the settings.

Table 1: Experimental configuration

Parameter	**Synthetic**	**Digits**
Observations	150	1,797
Features	2	64
Clusters (K)	3	10
Trials per method	20	20
Seeds	0–19	0–19
Max. iterations	100	100
Tolerance	10^{-4}	10^{-4}
Distance	Euclidean	Euclidean

4.3 Repeated Trials

Each initialization method was evaluated 20 times per dataset, using seeds from 0 to 19.

The synthetic observations were generated once using seed 42 and remained unchanged throughout the repeated trials.

Within each trial, both methods received the same dataset and random seed, but selected their initial centroids according to different strategies.

The final inertia was recorded for every run. Mean inertia, standard deviation, and best observed inertia were calculated separately for both methods.

5 Results

5.1 Synthetic Dataset Results

Table 2 summarizes the 20-trial results for the synthetic dataset.

Table 2: Synthetic dataset: 20 trials

Metric	**Random**	**K-means++**
Mean inertia	329.43	286.18
Std. deviation	188.48	0.00
Best inertia	286.18	286.18

K-means++ achieved a final inertia of approximately 286.18 in all 20 trials.

Random initialization reached the same low inertia in 19 trials, but produced one unfavorable result.

In trial 2, corresponding to seed 1, random K-means converged to an inertia of approximately 1151.01, compared with 286.18 for K-means++.

This single poor result increased the random method's mean inertia and standard deviation.

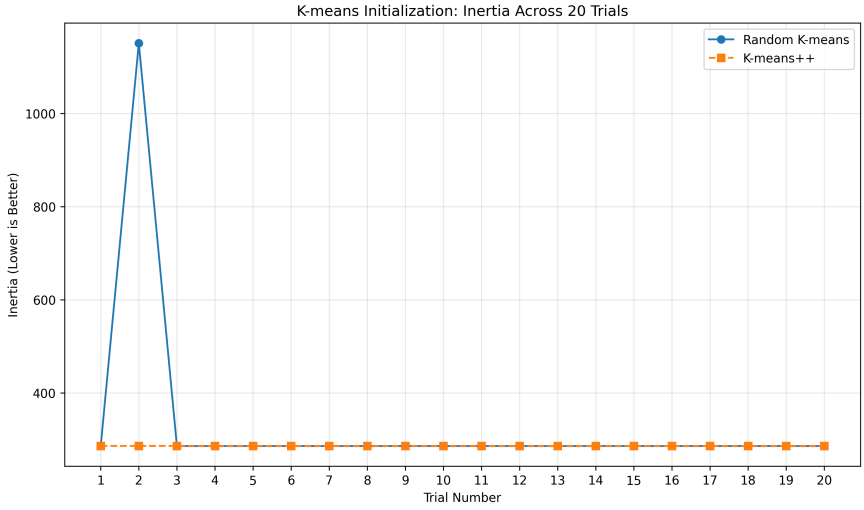


Figure 3: Synthetic dataset: inertia across 20 trials for both methods.

Figure 3 illustrates the single large increase in random-initialization inertia.

The final cluster assignments from that trial were also compared.

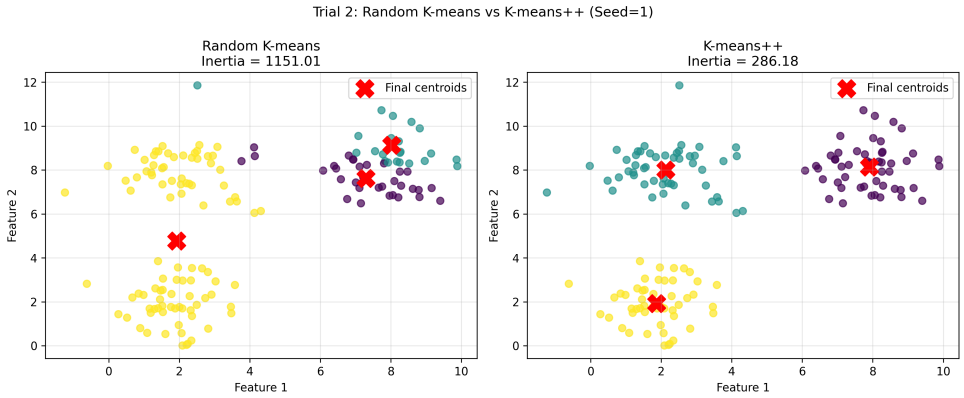


Figure 4: Trial 2 (seed 1): final clustering solutions with random initialization and K-means++.

The visualization shows that random initialization produced a partition that did not correspond well to the three visible Gaussian groups.

K-means++ recovered the expected group structure in this trial.

5.2 Handwritten Digits Results

Table 3 reports the results obtained from 20 trials with $K = 10$.

Table 3: Handwritten digits: 20 trials

Metric	**Random**	**K-means++**
Mean inertia	1,178,338.22	1,179,855.95
Std. deviation	19,366.72	16,500.79
Best inertia	1,165,160.38	1,165,372.07

Unlike the synthetic experiment, K-means++ did not achieve the lower mean inertia on Digits.

Random initialization achieved a mean inertia of 1,178,338.22, compared with 1,179,855.95 for K-means++.

The difference is approximately 0.13% relative to the random method's mean inertia.

Random initialization also achieved a slightly lower best observed inertia.

However, K-means++ produced a smaller standard deviation: 16,500.79 compared with 19,366.72 for random initialization.

This represents an approximately 14.8% reduction in observed standard deviation.

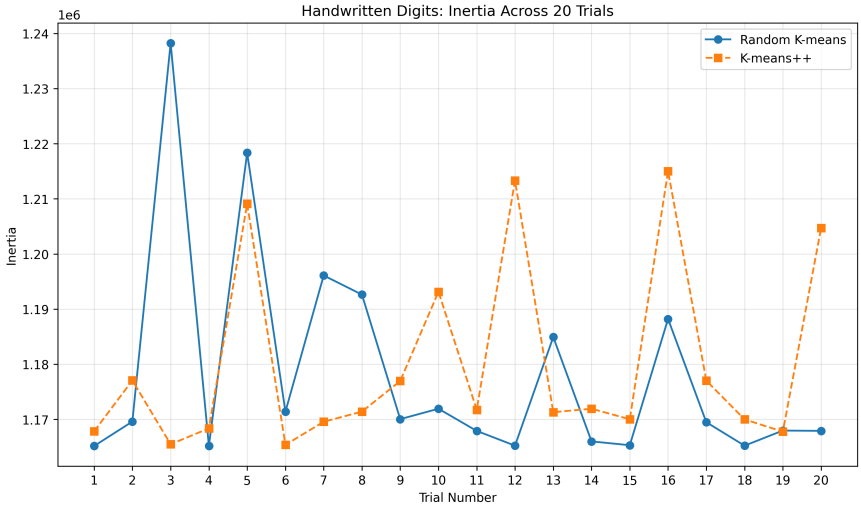


Figure 5: Handwritten digits: inertia across 20 trials.

Both methods produced different inertia values across the random seeds. K-means++ was somewhat more consistent, although its average inertia was slightly higher in this experiment.

6 Discussion

The experiments demonstrate that centroid initialization can affect both the quality and consistency of K-means clustering. However, the effect was not identical across the two datasets.

On the synthetic Gaussian dataset, K-means++ produced the same low inertia in every tested trial. Random initialization achieved the same result in most trials, but one unfavorable initialization led to a much larger clustering cost.

This illustrates an important advantage of careful seeding: it can reduce the likelihood of starting configurations that lead to poor local solutions.

Nevertheless, the synthetic results should be interpreted carefully. The higher mean inertia of random initialization was driven by a single outlier. In 19 of the 20 trials, both methods achieved the same result.

On the handwritten digits dataset, the advantage of K-means++ was less clear. Random initialization achieved slightly lower mean and best inertia, while K-means++ had lower variability.

One possible explanation is that the synthetic dataset contains three well-separated groups, where spreading initial centroids across the feature space is particularly effective.

The handwritten digits dataset has a more complex structure, with 64 features and potentially overlapping groups. In such settings, widely separated initial centroids do not necessarily lead to the lowest final clustering objective.

These findings are compatible with the theoretical motivation of K-means++ [1]. A guarantee concerning expected initialization cost does not guarantee a better final solution for every dataset or seed.

Several limitations should also be considered. The experiments used only two datasets and 20 random seeds per method. Therefore, the observations should not be generalized to all clustering problems.

Furthermore, the evaluation focused on inertia rather than external clustering quality metrics. A lower inertia does not necessarily mean that the resulting clusters correspond more closely to the true digit classes.

Overall, the results suggest that K-means++ is a useful initialization strategy, especially when reducing sensitivity to poor starting centroids is important. However, empirical performance still depends on the dataset.

7 Conclusion

This project compared random initialization and K-means++ for K-means clustering using two datasets and 20 repeated trials per method.

On the synthetic Gaussian dataset, K-means++ achieved consistent low inertia across all trials, while random initialization occasionally converged to a substantially worse solution.

On handwritten digits, K-means++ showed lower variability, but random initialization achieved slightly better mean and minimum inertia.

The main conclusion is that careful centroid initialization can improve the reliability of K-means, but it does not guarantee better final inertia in every experimental setting.

Future work could investigate additional datasets, larger numbers of trials, computational runtime, and clustering quality metrics beyond inertia.

References

[1] Arthur, D. and Vassilvitskii, S. (2007). *k-means++: The Advantages of Careful Seeding*. Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 1027–1035.

[2] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825–2830.

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